

HEALTHCARE ANALYTICS · SYNTHETIC DATA

Time-to-Attribution: Estimating Days from Coverage Start to First PCP Assignment

A statistical analysis of how long it takes a new health plan member to be attributed to a Primary Care Provider after coverage begins — and which member segments attribute fastest or slowest.

34 MEDIAN DAYS TO ATTRIBUTION	86.7% OVERALL ATTRIBUTION RATE	73.9% ATTRIBUTED WITHIN 60 DAYS	5,000 SYNTHETIC MEMBER RECORDS
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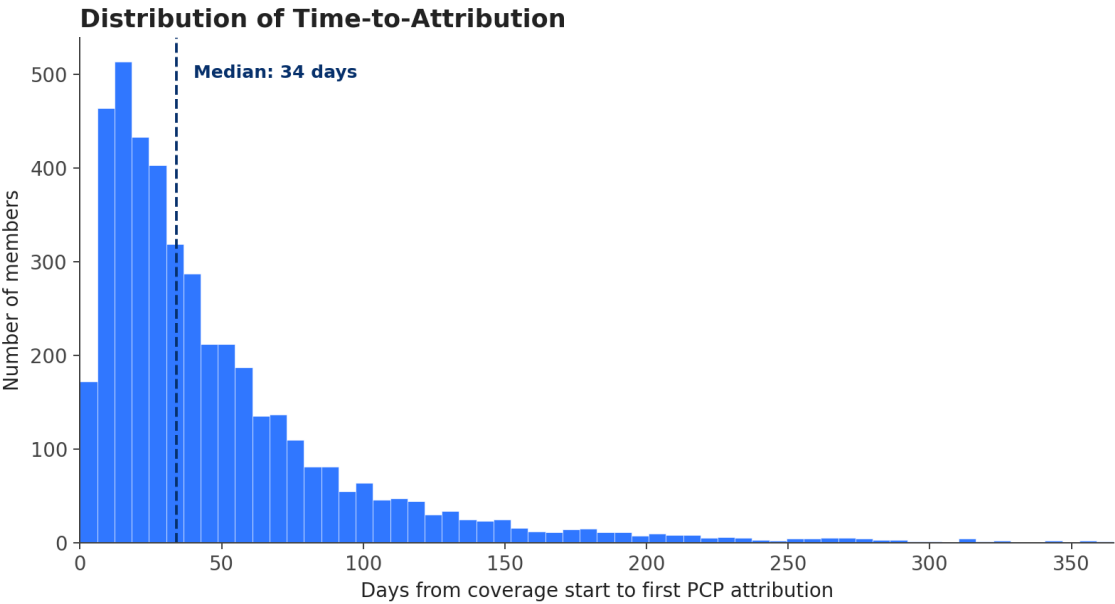
THE PROBLEM

Health plans use claims-based logic to attribute members to a Primary Care Provider (PCP), typically based on visit patterns within a rolling window (6–24 months). The speed of that attribution matters: members who go unattributed for long stretches can fall outside care coordination programs and may be miscounted in quality measures like HEDIS. This analysis asks a simple operational question: how long does attribution actually take, and for whom does it take longest?

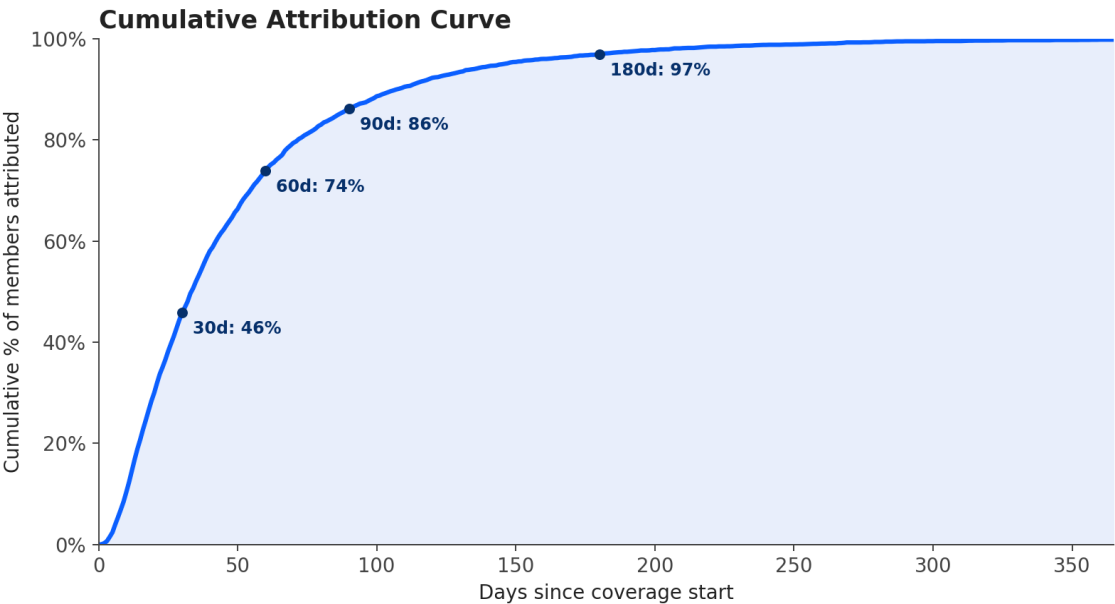
APPROACH

- Built a synthetic dataset of 5,000 members with realistic coverage start dates, plan types, regions, and age groups
- Modeled days-to-attribution as a right-skewed distribution adjusted by known real-world drivers: plan type, region, and age
- Calculated cumulative attribution rates at 30/60/90/180-day milestones and segmented results by plan type and region
- Built all charts in Python (pandas, numpy, matplotlib) — full script available on request

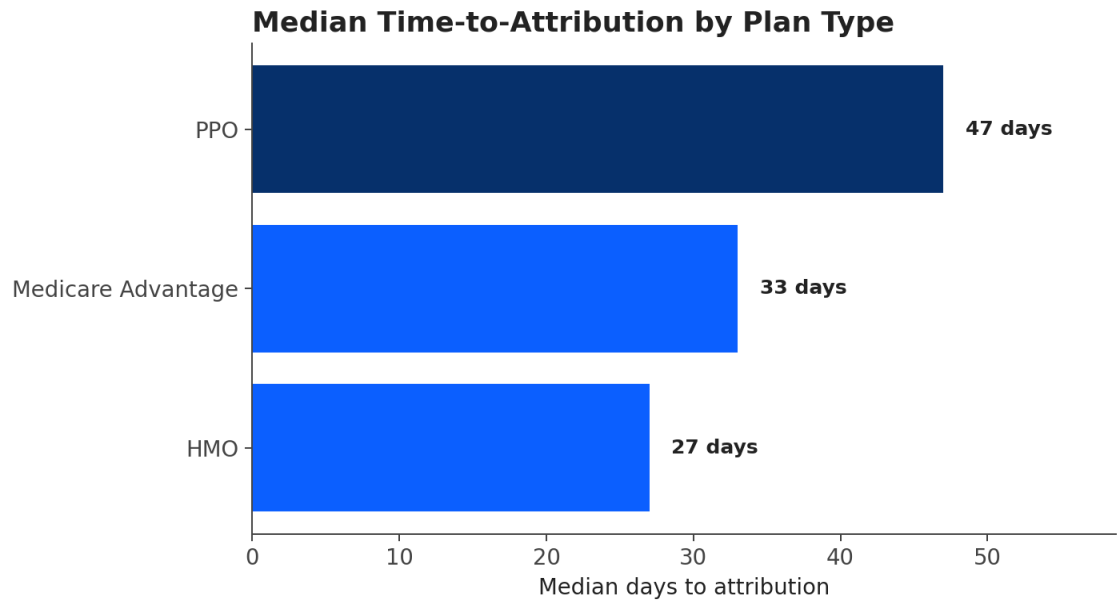
FINDINGS



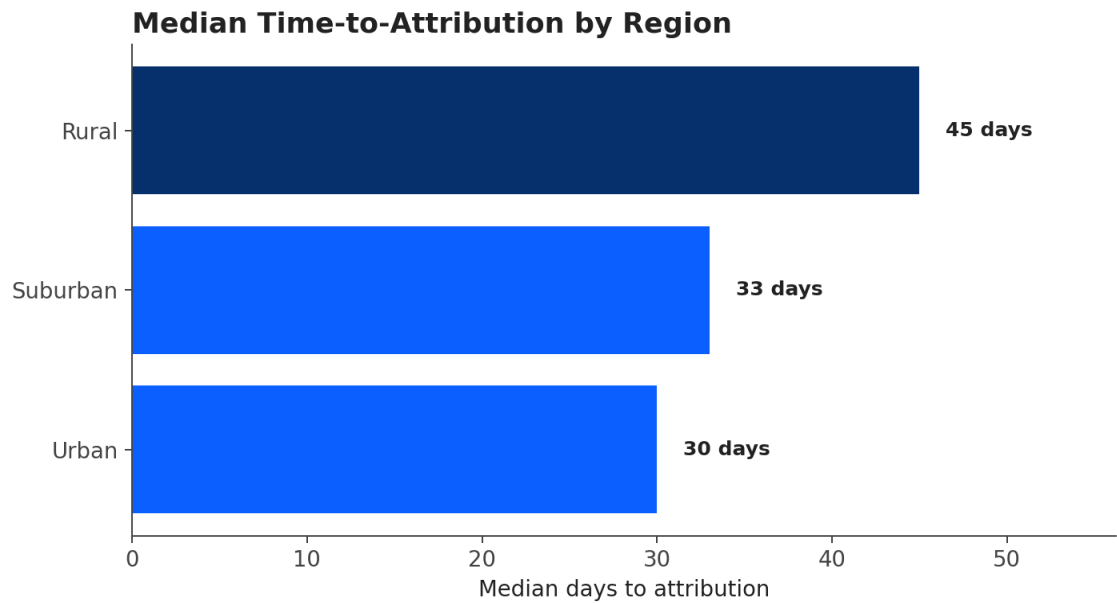
Most members who attribute do so quickly — the distribution is heavily right-skewed with a median of 34 days, but a long tail extends past 200 days.



Nearly half of members are attributed within 30 days, and the curve flattens after ~120 days — suggesting interventions should target the first month.



PPO members attribute notably slower than HMO members, consistent with fewer enrollment-time PCP requirements.



Rural members attribute slower than urban members, consistent with fewer provider options.

TAKEAWAY

If a health plan wants to shorten time-to-attribution, the highest-leverage levers are plan design (encouraging or requiring PCP selection at enrollment) and provider network access in rural areas — not just faster claims processing. This kind of segment-level view turns a single average into a set of targeted, actionable interventions.

Note on data: This project is inspired by real-world member attribution work in health plan analytics, but all data shown here is fully synthetic — generated programmatically to illustrate the method. No real member, claims, or employer data was used or referenced.